

Q1. State True/False and justify with a short answer. (Binary marking. Both the statement and the reason should be correct. 1 mark each)

- (a) Bandwidth of the digitally modulated signal can be estimated from the energy spectral density of the linearly modulated signal $u(t)$.
 - (b) Bandwidth of the digitally modulated signal does not depend on the Fourier transform of the message signal.
 - (c) Coherent demodulation for FSK is possible for $\Delta f = 5/T$.
 - (d) Rectangular pulse suffers from the issue of ISI in the presence of a timing offset of $\leq 0.5T$, where T is the width of the rectangular pulse.
 - (e) A cosine signal with period T can be used as a Nyquist pulse, as it gives zeros at intervals of $T/2$.
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Q2 Suppose that white noise $n(t)$ with PSD $\sigma^2 = \frac{N_0}{2} = \frac{1}{4}$ is passed through an LTI system with impulse response $g(t)$ to obtain the output $y(t) = n(t) * g(t)$. Consider $g(t)$ to be the triangle shown in the figure below. Answer the following questions (5 marks):

- (a) Find the autocorrelation function and PSD of y .
- (b) Are $y(5)$ and $y(6.5)$ independent random variables?
- (c) Are $y(5)$ and $y(8.1)$ independent random variables?
- (d) Is y a stationary random process?

